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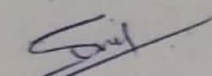
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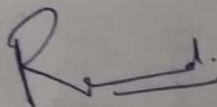
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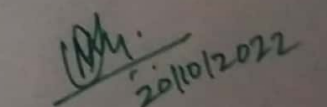
References:

Chandrika Prasad, Advance Engineering Mathematics, Stanford Publishers and Distributors

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20/10/2022
DEAN ACDMICS

UNIT-1 (ADVANCE ENGINEERING MATHEMATICS)

HISTORICAL DEVELOPMENT OF OPTIMIZATION

LECTURE NO: -18

Historical Development of Optimization:

Optimisation is the act of obtaining the best result under given circumstances in design, construction and maintenance of any engineering system, engineers have to take many technological and managerial decisions at several stages. The ultimate goal of all such decisions is either minimise the effect required or the desired benefit. Optimisation can be defined as the process of finding the conditions that give the maximum and minimum value of a function.

Need of Optimisation

The search for extremes inspires mountaineers, scientists, mathematician and the rest of the human race. Every new generation of computers allows for attacking new types of problems and calls for new methods. The goal of the theory is the creation of reliable methods to catch the extremum of a function by an intelligent arrangement of its evaluations. This theory is vitally important for modern engineering and planning that incorporates Optimisation at every step of the complicated decision making process. In engineering, every engineer tries to create the best project and relies on Optimisation methods to achieve the goal.

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Recently, the author conducted research to examine the status of technology education regarding the infusion of engineering design concepts (Kelley, 2008). Participants from this study revealed that technology education curriculum content currently does not emphasize optimization techniques as a part of the engineering design process. One of lowest-ranking survey items for time per typical use was the item: use optimization techniques to determine optimum solutions to problems, mean of 1.82 using a 5-point Likert scale. In the author's search to understand why technology educators have not emphasized this phase of the engineering design process to a greater degree, the author discovered that there was very little in engineering design textbooks or engineering design curriculum at the secondary level regarding optimization. One of the few pre-engineering sources that broached the subject, Engineering Your Future, (Gomez, Oakes, & Leone, 2006) dedicated only one page to optimization. However, the members of the National Center for Engineering and Technology Education (NCETE) have identified optimization as an important concept in engineering design (Merrill, Custer, Daugherty, Westrick, & Zeng, 2007). NCETE members have identified three specific core design concepts important in the engineering design process and have termed

these concepts Constraints, Optimization, and Predictive Analysis or COPA. The COPA concept has been used to help technology education teachers quickly identify the core concepts of engineering design. NCETE has developed some activities designed to deliver COPA in technology education. This article will focus specifically on Optimization as a way to inform technology educators about the concept and provide an example of optimization through the case of Charles Lindbergh's famous 1929 flight from New York to Paris. The author will present a historical case using Lindbergh's own words from his biography, *The Spirit of St. Louis* (1953), to illustrate that he used an optimization process to make design decisions about his plane and flight, which led to his success much more than just being "Lucky Lindy."

Lindbergh and The Spirit of St Louis: An Example in Optimization When Lindbergh set out to win the Orteg prize for being the first aviator to fly nonstop from New York to Paris, the aviation technology was available to accomplish such a goal. However, other aviators more experienced than Lindbergh (Byrd, Davis and Wooster, Fokker, and Nungesser) attempted the nonstop flight, resulting in crashes at take off or losses at sea (Lindbergh, 1953). These failures did not occur because the famous flyers lacked the advanced technology of the time or because they were unskilled fliers. Money was not an issue: aviators such as Byrd, Fonck, Davis, and Nungesser poured thousands of dollars into multiple-engine airplanes, some of which never lifted off the ground. Why was Lindbergh successful? He optimized for the best available solution (*The Spirit of St Louis*) under the given constraints and conditions—a technique Lindbergh learned as an engineering student at the University of Wisconsin. There were many issues for Charles Lindbergh to consider as he planned for his nonstop transcontinental flight. At the forefront of all of his concerns was, of course, his safety. Lindbergh (1953) writes: "Safety at the start of my flight means holding down weight for the take off. Safety during my flight requires plenty of emergency equipment. Safety at the end of my flight demands ample reserve of fuel. It is impossible to increase safety at one point without detracting from it at another. I must weigh all these elements in my mind, and attempt to strike some balance" (p.97). What Lindbergh illustrates through these words is that to engineer anything requires decision making and balancing constraints and criteria to implement the best possible solution. Let's review the final decisions that Lindbergh made for the design of his aircraft and the plans of his flight assessed against the constraints and criteria he listed above: (1) keeping weight down, (2) safety during flight, and (3) ample reserve of fuel. The first constraint, keeping weight of the plane down, directly correlates to the design criteria ample reserve of fuel. In order to keep weight down, Lindbergh's first design choice was something the other aviators never considered, a single-engine monoplane. When challenged by financial backers to consider a multiengine plane, Lindbergh responded, "I'm not sure three engines would really add much to safety on a flight like that (over water). There'd be three times the chance of engine failure; and if one of them stopped over the ocean, you probably couldn't get back to land with the other two. A multiengine plane is awfully big and heavy" (Lindbergh, 1953, p.26). Lindbergh also chose a monoplane over a biplane. When asked about this decision, Lindbergh said, "it [monoplane] is more efficient than a biplane, there's more room in the wing for gasoline, and it can carry more ice (on the wing)" (p. 103). Another decision made by Lindbergh to keep the weight down was to fly the plane solo. Clearly the greatest risk in Lindbergh's plan was flying solo for over 33 hours and 30 minutes. However, Lindbergh believed flying alone was his greatest asset. "By

flying alone, I've gained in range, in time, in flexibility; and above all, I've gained in freedom" (p. 192). By flying alone, Lindbergh was able to add more weight in the form of additional fuel necessary to make the transatlantic flight and ensure that he had a safety cushion of extra fuel in case of a navigational error or if forced to turn back due to inclement weather. This decision addressed all the major constraints and criteria: (1) keeping weight down, (2) safety during flight, and (3) ample reserve of fuel. Lindbergh made some decisions about what to carry and, more specifically, what not to carry, that might cause some to wonder if he had carefully considered his own safety during flight. For example, Lindbergh chose not to carry an aircraft radio, a parachute, or a sextant (tool for navigation). These items seem necessary for a pilot's safety. However, through the optimization process, Lindbergh rationalized that these items would cost more in added weight to the plane than they would be worth in practical usage. The parachute was a tough item to reject; however, Lindbergh provides a logical rationale for not carrying one. "I considered carrying a parachute, but decided against it. A parachute would have cost twenty pounds—a third of an hour of fuel—enough food and water for many days" (pp. 212-213). Lindbergh also supported his decision to not carry a parachute with the rationale that he could only use a parachute in one part of the journey (over Nova Scotia). Lindbergh's flight pattern had him flying too low for a parachute (over Long Island and New England), over water, or (over Europe) when the plane would be light enough for a safe stall-landing. Lindbergh chose not to carry a naval radio because, at the time, these instruments were very heavy, difficult to use, and were unreliable. Lindbergh wrote: "I find that naval radios are much too heavy for my single-engine plane, and that their value on a flight like mine is doubtful" (p. 96). He addresses the issue of not carrying a sextant when he wrote: "I couldn't possibly use a sextant . . . I couldn't take a sight and fly the plane at the same time. The slightest turn throws the bubble off. The Spirit of St Louis won't hold a straight course for two seconds by itself. Besides, there's the weight—you can't carry everything on a record flight" (p.237). "If we'd tried to carry every safeguard, the plane couldn't have gotten off the ground—dump valves, parachute, radio, sextant" (p.237). In fact, that is certainly one reason why Byrd, Fonck, Davis, and Nungesser crashed on takeoff; they all tried to carry more weight than the plane could handle. Lindbergh rationalized these decisions by determining what he gained by giving up these items. He wrote: "We'll trade radio and sextant weight for extra gasoline. What I lose in navigational accuracy, I hope to gain twice over in total range" (p. 96). Lindbergh did choose to carry some items for personal safety, including a rubber boat, red flares, emergency food rations, and an extra gallon of drinking water. In all, these items weighed over 30 lbs; equivalent to over a half an hour of flying time in fuel weight. These items, Lindbergh considered, were optimal options for the conditions he would be under during his flight. Lindbergh had aeronautical engineers working with him, as Ryan Air custom designed and built The Spirit of St. Louis. Lindbergh made final decisions on how the plane was designed. One unique feature of the aircraft was the location of the cockpit behind the gas tank. Lindbergh believed that locating the cockpit behind the gas tank gave him a better chance in case of a crash landing. This was a very abnormal design that placed the gas tank in the view of the pilot. That fact was no issue to Lindbergh; he decided to design the plane without a windshield. He writes: "There is not much need to see ahead in normal flight. I won't be following any airways . . . All I need is a window on each side to see out through. The top of the fuselage could be the top of the cockpit. A cockpit like that wouldn't add any resistance at all" (p.87).

Remember, Lindbergh, like all good engineers, made decisions based upon defined criteria and identified constraints. In this case, Lindbergh, in his own words writes, "I think we ought to give first consideration to efficiency in flight; second to protection in a crack-up; third, to pilot comfort (p. 99). Lindbergh also had to consider keeping costs down, so he chose to keep the design of the plane very

Historical Development

The existence of Optimisation methods can be traced to the days of Newton, Lagrange and Cauchy. The development of differential calculus methods of Optimisation was possible because of the the contribution of Newton and Leibnitz to calculus. The foundation of Calculus variations which deals with the minimization of functional were laid by Bernoulli, Euler, Lagrange and Weierstrass.

The history dates back to late 30's and early 40 's when military commands of US and UK, during Second World War formed a committee of scientists to design for strategic and tactical military operation by doing practical research confined to the available resources was to fight the war to win with maximum damage to the enemy.

Inspired by the the success of military in the Second World War, the industry people were attracted towards the team of scientists. The motive behind this war to design the algorithms that can maximize the cost of production and maximize the profit there of.

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

ENGINEERING APPLICATIONS OF OPTIMIZATION

LECTURE NO: -19

Engineering Applications of Optimization:

Optimization, in its broadest senses, can be applied to solve any engineering problem, to indicate the wide scope of the subject, some typical applications from different engineering disciplines are given optimization theory finds ready applications in all branches in engineering in the four primary areas.

- 1.Design of components or entire system
2. Planning and analysis of existing system
3. Engineering analysis and data reduction
4. control of dynamic system

1. Design optimization

(i) Design of aircraft and aerospace structures for minimum weight

(ii) Design of civil engineering structure such as frames foundation, bridges, towers, chimneys and dams for minimum cost.

(ii) Optimal plastic design of structures

(iv) Optimum design of linkages, comes, gears, machines tools and others mechanical components.

(v) Minimum- weight design of structures for earthquake, wind and other types of random loading.

(vi) Design of material handling equipment such as conveyors, truck and cranes for minimum fast.

(vii) Design of pumps, turbines and heat transfer equipment of maximum efficiency.

(viii) Optimum design of electrical machinery such as motors, generators and transformers.

(ix) Optimum design of electrical networks.

(x) Optimum design of chemical process equipment and plant.

(ix) Design of optimum pipeline networks for process.

(iix) Optimum design of control system.

2. Planning and Analysis

- (i) Maximizing benefits or minimizing product cost in various manufacturing and construction process.
- (ii) Optimal path finding in road network/freight handling process.
- (iii) Optimal allocation of resources or services among several activities to maximizing the benefits.
- (iv) Optimal production planning, controlling and scheduling.
- (v) Feeding strategies for maximize gross margin in pig production.
- (vi) Shortest path algorithm are applied to automatically find direction between physical location, such as driving direction on web mapping websites like map query or google maps.
- (vii) Planning the best strategy to obtain maximum profit in the presence of a competitors.
- (viii) planning of maintenance and replacement of equipment to reduce operating costs.
- (ix) Planning of Shortest rout taken by a salesperson visiting various cities during one hour.

3, Engineering Analysis

- (i) Finding the optimal trajectories of space vehicles minimization of total cost per component in selecting the optimal machinery parameters feed and speed in matel cutting.
- (ii) Optimal producing planning, controlling and scheduling.

4. Control of Dynamic System

- (i) Determination of the equilibrium composition of a chemical mixture that minimizes gibbs free energy of the system.
- (ii) Determination of the steady state current flows in an electrical networks minimizing the I^2R power loss subject to a set of linear constraints.

5. Other Engineering Applications

- (i) Inventory control
- (ii) Queuing problem
- (i) Optimizing the cost of transportation of man material of a big projects.
- (ii) Optimum methods of transmission and distribution of electricity.

Basic Components of an Optimization Problem

Optimization problem are made up of three basics ingredients

1. Objective Function

An objective function which we want to minimize the cost. In fitting experimental data to a user defined model, we might minimize the total deviation of observed data from

prediction based on the modal. In designing and automobile panel, we might want to minimize the strength.

2. Variables

The set of unknown or variables which affects the value of the objective function. In the manufacturing problems the variables might include the amount of different resources used or the time spent on each activity. In fitting the data problem, the unknowns are the parameters that define the model. In the panel design problem, the variables used, define the shape and dimensions of the panel.

3. Constraints

The set of constraints allow the unknown to take on certain values but exclude others. For the manufacturing problem, it does not make sense to spend a negative amount of time on any activity, so we certain all the "time" variables to be non-negative. In the panel design problem, we could probably want to limit the weight of the product and to constraint shape.

A mathematically programming or an optimization problem can be started as follows:

Minimizes $f(x)$, where

$X = [x_1, x_2, \dots, x_n]$

Subject to the constraints

$G_j(x) \leq 0 \quad j=1,2,3,\dots,m$

Where x is an n - dimensional vector known as design vector, $f(x)$ is the objective function and $g_j(x)$ are known as inequality constraints the number of variables (n) and number of constraints (m and or p) need not be related in any way. The problem given in equation is called constrained optimization problem.

References:

Engineering Mathematics, T Veerarajan, TMH

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

SINGLE VARIABLE OPTIMIZATION WITH AND WITHOUT CONSTRAINTS

LECTURE NO: -20

Single Variable Optimization With and Without Constraints:

The classical optimization techniques are useful in find the optimum solution or unconstrained maxima or minima of continuous and differentiable functions. These are analytical methods and make use of differential calculus in locating the optimum solution. The classical methods have limited scope in practical applications as some of them involve objective functions which are not continuous and /or differentiable. Yet the study of these classical techniques of optimization form a basis for developing most of the numerical techniques that have evolved into advanced techniques more suitable to today's practical problems. These methods assume that the function is differentiable twice with respect to the design variables and that the derivatives are continuous.

Three main types of problems can be handled by the classical optimization techniques:

1. Single variable Functions
2. Multivariable functions with no constraints and
3. Multivariable functions with both equality and inequality constraints

For problems with equality constraints the Lagrange Multiplier method can be used. If the problem has inequality constraint, the Kuhn-Tucker conditions can be used to identify the optimum solution. These methods lead to a set of non-linear simultaneous equations that may be difficult to solve.

Unconstrained Optimization:

Single Variable Optimization:

(A) Stationary Points: For a continuous and differentiable function $f(x)$, a stationary point x^*

is a point at which the function vanishes. $f'(x)=0$ at $x=x^*$. x^* belongs to its domain of definition.

A stationary point may be minimum, maximum or an inflexion point.

Local or Global Optimum:

A function is said to have a relative or local minimum at $x=x^*$ if $f(x^*) \leq f(x)$, $x=x^* \pm h$ for all sufficiently small positive and negative values of h , in the near vicinity of the point x .

Similarly a point x^* is called relative or local maxima if $f(x^*) \geq f(x)$, $x=x^* \pm h$ for all values of h sufficiently close to zero.

A function is said to have a global or absolute minimum at $x=x^*$ if $f(x^*) \leq f(x)$ for all x in the domain over which $f(x)$ is defined. Similarly a function is said to have a global or absolute maximum at $x=x^*$ if $f(x^*) \geq f(x)$ for all x in the domain over which $f(x)$ is defined.

Function of a single variable:

Consider the function $f(x)$ defined for $a \leq x \leq b$. To find the value of $x^* \in [a, b]$ such that x^* maximize $f(x)$, we need to solve a single variable optimization problem.

Necessary Condition:

For a single variable function $f(x)$ defined for x which has a relative maximum at $x=x^*$ and if the derivative $f'(x)=d/dx(f(x))$ exist as a finite number at $x=x^*$ then $f'(x^*)=0$

Sufficient Condition:

For the same function stated let $f'(x^*) = f''(x^*) = \dots = f^{n-1}(x^*) = 0$, but

$f^n(x^*) \neq 0$ then it can be said that $f(x^*)$ is:

- (a) A minimum value of $f(x)$ if $f^n(x^*) > 0$ and n is even
- (b) A maximum value of $f(x)$ if $f^n(x^*) < 0$ and n is even
- (c) Neither a maximum nor minimum if n is odd, this is a point of inflexion.

1. Find the optimum value of the function $f(x) = x^2 + 3x - 5$. Also state if the function attains a maximum or the minimum.

Sol. For maxima or minima

$$F'(x) = 2x + 3$$

$$x^* = -3/2$$

$F''(x) = 2$ which is positive, hence the point $x^* = -3/2$

Is the point of minima and the function attains the minimum value of $-29/4$ at this point.

2. Find the optimum value of the function $f(x) = (x-2)^4$ and also state that if the function attains a maximum or a minimum at $x=2$.

Sol. $F(x) = (x-2)^4$

$$F'(x) = 4(x-2)^3 \text{ or } x = x^* = 2$$

For maxima and minima,

$$F''(x^*) = 12(x^* - 2)^2 = 0 \text{ at } x^* = 2$$

$$F'''(x^*) = 24(x^* - 2) = 0$$

$$F^{iv}(x^*) = 24 \text{ at } x^* = 2$$

Hence $f^n(x)$ is positive and $n = 4$ is even hence the point $x = x^* = 2$ is the point of minimum and the function attains a minimum value of 0 at this point.

3. Analyze the function and classify the stationary point as maxima, minima and point of inflexion. $F(x) = 2x^5 - 45x^4 + 40x^3 + 5$.

Sol: $f'(x) = 60x^4 - 180x^3 + 120x^2 = 0$

$$x^4 - 3x^3 + 2x^2 = 0$$

$$x = 0, 1, 2$$

- (i) Consider the point $x=0$ at

$$f''(0) = 240(0)^3 - 540(0)^2 + 240(0) = 0 \text{ at } x=0$$

$$f'''(0) = 720(0)^2 - 1080(0) + 240 = 240 \text{ at } x=0$$

Since the third derivative is non-zero, $x=0$ is neither a point of maximum or minimum be it is a point of inflexion.

- (ii) Consider at $x=2$

$$f''(1) = 240(1)^3 - 540(1)^2 + 240(1) = -60 \text{ at } x=1$$

Since the second derivative is positive, the point $x=2$ is a point of local maxima with a maximum value $f(x)=12$

- (iii) Consider $x=2$

$$f''(2) = 240(2)^3 - 540(2)^2 + 240(2) = 240 \text{ at } x=2$$

Since the second derivative is positive, the point $x=2$ is a local minima with minimum value of $f(x)=-11$

4. Prove that $x/1+x.\tan x$ is a maximum when x is given by the equation $x=\cos x$

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors

UNIT-3 (ADVANCED ENGINEERING MATHEMATICS)

MULTIVARIABLE OPTIMIZATION WITHOUT CONSTRAINTS

LECTURE NO: -21

Multivariable Optimization without Constraints

Hessian Matrix: The Hessian matrix is the square matrix of second order partial derivatives of a function. Given the real-valued function $f(x_1, x_2, \dots, x_n)$ if all second order partial derivatives of f exist, then the Hessian matrix of f is the matrix

$$H(f)_{ij}(x) = D_i D_j f(x) = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

Where $x = (x_1, x_2, \dots, x_n)$ and $D_i = \frac{\partial}{\partial x_i}$ is the differentiation operator with respect to the i^{th} argument and the Hessian becomes

For two variable function $f(x, y)$

$$H_f(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$$

Some Definition:

Suppose that we have a square matrix M of order n . then M is:

- (i) **Positive Definite:** if the determinants of all of its principal submatrix are all positive or eigen value of M are all positive real number.
- (ii) **Negative Definite :** If the determinants of all its leading principal submatrix are non-zero and alternate in sign with first ($k=n$) being negative or the eigen values of M are all negative or the eigen of M are all negative real number.

If we happen to have a few zero determinants or zero eigen value, then we can say that M is:

- (iii) **Positive Semi-definite:** If the determinants of all its principal submatrices are all non-negative real numbers.
- (iv) **Negative Semi-definite:** If the determinants of all of its principal submatrices alternate in sign, starting with a negative or all of its eigen values are non-positive real numbers.

Necessary Conditions for extreme point:

If $f(x)$ has an extreme point (maximum or minimum) at $x = x^*$ and if the first partial derivatives of $f(x)$ exist at x^* , then

$$\frac{\partial}{\partial x_1} f(x^*) = \frac{\partial}{\partial x_2} f(x^*) = \dots = \frac{\partial}{\partial x_n} f(x^*) = 0$$

Sufficient Condition for Extreme Point:

A sufficient condition for a stationary point x^* to be an extreme point is that the matrix of second partial derivatives (Hessian matrix) of $f(x)$ evaluated at x^* is

- (i) Positive definite when x^* is a relative minimum point.
- (ii) Negative definite when x^* is a relative maximum point.

$$J|_{x=x^*} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{x=x^*}$$

is the matrix of second partial derivatives and is called the Hessian matrix of $f(x)$.

Thus x^* is a relative minimum of $f(x)$ if $J|_{x=x^*}$ is positive definite and x^* is a relative maximum of $f(x)$ if $J|_{x=x^*}$ is negative definite.

Semidefinite case: For the sufficient condition, if the Hessian matrix is semi-definite, we investigate the higher order derivatives in the Taylor series expansion. However, the algebra becomes a bit complicated and we usually avoid it.

Saddle point:

If the Hessian matrix of a function $f(x,y)$ at a point (x^*, y^*) at which $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$

is neither positive definite nor negative definite, then point (x^*, y^*) is called a saddle point.

1. The profit per acre of a farm is given by

$$F(x,y) = 20x + 26y + 4xy + 4x^2 - 3y^2$$

Where x and y denote, respectively, the labour cost and the fertilizer cost. Find the value of x and y to maximize the profit.

Sol: The necessary conditions for extreme point are

$$\frac{\partial f}{\partial x} = 20 + 4y - 8x = 0$$

$$\frac{\partial f}{\partial y} = 26 + 4x - 6y = 0$$

This gives $x=7, y=9$

The sufficient condition

$$J(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} -8 & 4 \\ 4 & -6 \end{vmatrix}$$

The determinants of the square submatrices of J are

$$J_1 = |-8| = -8$$

$$J_2 = \begin{vmatrix} -8 & 4 \\ 4 & -6 \end{vmatrix} = 32$$

Thus the matrix J is negative definite and hence $(7, 9)$ corresponds to the maximum of profit of $f(x,y)$.

1. Determine the nature(positive definite, negative definite or indefinite) of the following matrices

$$(i) \quad A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{pmatrix}$$

$$\text{Sol: } A_1 = |3| = 3$$

$$A_2 = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} = 8$$

$$A_3 = \begin{vmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{vmatrix} = 36$$

So A is positive definite

2. Find the extreme points of the function

$$F(x, y, z) = x^2 + 4y^2 + 4z^2 + 4xy + 4xz + 16yz$$

$$\text{Sol: } \frac{\partial f}{\partial x} = 2x + 4y + 4z = 2(x + 2y + 2z)$$

$$\frac{\partial f}{\partial y} = 8y + 4x + 16z = 4(x + 2y + 4z)$$

$$\frac{\partial f}{\partial z} = 8z + 4x + 16z = 4(x + 4y + 2z)$$

$$\text{At an extreme point } \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = 0$$

$$(x + 2y + 2z) = 0$$

$$(x + 2y + 4z) = 0$$

$$(x + 4y + 2z) = 0$$

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 4 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

By Gauss elimination method

$$\begin{pmatrix} 1 & 2 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$(x, y, z) = (0, 0, 0)$ is the unique solution.

Hessian matrix J is given by

$$J = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial z \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 4 & 8 & 16 \\ 4 & 16 & 8 \end{bmatrix}$$

$$J_1 = |2| = 2$$

$$J_2 = \begin{vmatrix} 2 & 4 \\ 4 & 8 \end{vmatrix} = 0$$

$$J_3 = \begin{vmatrix} 2 & 4 & 4 \\ 4 & 8 & 16 \\ 4 & 16 & 8 \end{vmatrix} = -128 < 0$$

So J is indefinite.

(0,0,0) is a saddle point.

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors

MITRC

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

MULTI-VARIABLE OPTIMIZATION WITH EQUALITY CONSTRAINTS

LECTURE NO: -22

Multi-Variable optimization with equality constraints

Method of lagrange multipliers:

To explain the method for general problem, we initially take two variable and one constraint consider.

Minimize $f(x_1, x_2)$

Subject to $g(x_1, x_2) = 0$

By the method of constrained variation, a necessary condition for the existence of an extreme

point (x_1^*, x_2^*) is
$$\left(\frac{\partial f}{\partial x_1} - \frac{\partial f}{\partial x_2} \cdot \frac{\partial g}{\partial x_1} \right) \bigg|_{(x_1^*, x_2^*)} = 0 \dots\dots\dots(1)$$

Provided $\frac{\partial g}{\partial x_2} \neq 0$

Define
$$\lambda = - \left(\frac{\partial f}{\partial x_2} \cdot \frac{\partial g}{\partial x_2} \right) \bigg|_{(x_1^*, x_2^*)} = 0 \dots\dots\dots(2)$$

Equation (1) can be written as.

$$\left(\frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1} \right) \bigg|_{(x_1^*, x_2^*)} = 0 \dots\dots\dots(3)$$

And equation (2) as.

$$\left(\frac{\partial f}{\partial x_2} + \lambda \frac{\partial g}{\partial x_2} \right) \bigg|_{(x_1^*, x_2^*)} = 0 \dots\dots\dots(4)$$

This λ so defined is called the lagrange multiplier in addition these two equation.

$$g(x_1, x_2) \bigg|_{(x_1^*, x_2^*)} = 0 \dots\dots\dots(5)$$

Thus equation (3), (4) and (5) represent the necessary condition for the point (x_1^*, x_2^*) to be an extreme point.

Construct the function.

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2)$$

By treating L as a function of variables x_1, x_2, λ the necessary condition for its extremum are given by

$$\frac{\partial L}{\partial x_1} = \frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1} = 0$$

$$\frac{\partial L}{\partial x_2} = \frac{\partial f}{\partial x_2} + \lambda \frac{\partial g}{\partial x_2} = 0$$

Necessary conditions for a general problem.

Consider the general problem.

Minimize $f(x)$

Subject to $g_j(x) = 0, j = 1, 2, \dots, m$

$$x = (x_1, x_2, \dots, x_n)^T, m < n$$

Necessary condition for the extreme point of L are.

$$\frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0 \quad i = 1, 2, 3, \dots, n$$

$$\frac{\partial L}{\partial x_j} = g_j(x) = 0 \quad j = 1, 2, 3, \dots, m$$

These are $(n+m)$ equation in $(n+m)$ unknowns. The solution of these equation gives.

$$x^* = \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{bmatrix} \quad \lambda^* = \begin{bmatrix} \lambda_1^* \\ \lambda_2^* \\ \vdots \\ \lambda_m^* \end{bmatrix}$$

The vector x^* gives the relative constrained minimum of $f(x)$ (sufficient conditions are to be verified,) while λ^* vector provides the sensitivity information.

Sufficient condition:

A sufficient condition for $f(x)$ to have a relative minimum at x^* is that the quadratic Q , defined by.

$$Q = \frac{\partial^n}{\partial x_i} \frac{\partial^n}{\partial x_j} dx_i dx_j$$

Evaluated at $x = x^*$ must be positive for all values of dx for which the constraints are satisfied.

The quadratic form be positive (negative) for all admissible variations dx is equivalent saying that each root of the polynomial in z , defined by the following detrimental equation, be positive (negative)

$$\begin{vmatrix} L_{11} - z & L_{12} & L_{13} \dots L_{1n} & g_{11} & g_{21} \dots g_{m1} \\ L_{21} & L_{22} - z & L_{23} & L_{2n} & g_{12} & g_{22} \dots g_{m2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ L_{n1} & L_{n2} & L_{n3} \dots L_{nn} & g_{1n} & g_{2n} \dots g_{mn} \\ g_{11} & g_{12} & g_{13} \dots g_{1n} & 0 & 0 \dots 0 \\ g_{21} & g_{22} & g_{23} \dots g_{2n} & 0 & 0 \dots 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ g_{m1} & g_{m2} & g_{m3} & g_{mn} & 0 & 0 & 0 \end{vmatrix} = 0 \dots (1)$$

$$\text{Where } L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j} (x^*, \lambda^*)$$

$$g_{ij} = \frac{\partial g_i}{\partial x_j} (x^*)$$

If roots of the eq. (1) are not all positive (negative), x^* do not represent an extreme point.

$$Q(1) \text{ minimize } f(x) = \frac{1}{2} (x_1^2 + x_2^2 + x_3^2)$$

$$\begin{aligned} \text{Subject to } g_1(x) &= x_1 - x_2 = 0 \\ g_2(x) &= x_1 - x_2 = 0 \\ g_3(x) &= x_1 + x_2 + x_3 - 1 = 0 \end{aligned}$$

By lagranges multiplier method.

Sol. Define the lagrange function.

$$L(x_1, x_2, x_3, \lambda, \lambda_2) = \frac{1}{2} (x_1^2 + x_2^2 + x_3^2) + \lambda_1 (x_1 - x_2) + \lambda_2 (x_1 + x_2 + x_3 - 1)$$

By necessary conditions

$$\left. \begin{aligned} \frac{\partial L}{\partial x_1} = 0 &\Rightarrow x_1 + \lambda_1 + \lambda_2 = 0 \\ \frac{\partial L}{\partial x_2} = 0 &\Rightarrow x_2 - \lambda_1 + \lambda_2 = 0 \end{aligned} \right\} \lambda_1 = \frac{x_2 - x_1}{2}; \lambda_2 = \frac{x_1 + x_2}{2}$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow x_3 + \lambda_2 = 0 \Rightarrow \lambda_2 = -x_3 = -\left(\frac{x_1 + x_2}{2}\right)$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow x_1 - x_2 = 0$$

$$\frac{\partial L}{\partial \lambda_2} = 0 \Rightarrow x_1 + x_2 + x_3 - 1 = 0$$

$$\therefore x_1^* = \frac{1}{3} = x_2^* = x_3^* \lambda_2^* = -\frac{1}{3}$$

To see, if $x_1^*, x_2^*, x_3^*, \lambda_1^*, \lambda_2^*$ really gives the minimum value we check the sufficient condition.

$$L_{11} = 1, L_{12} = L_{21} = 0, L_{13} = L_{31} = 0$$

$$L_{23} = L_{32} = 0, L_{22} = 1, L_{33} = 1$$

$$g_{11} = 1, g_{12} = -1, g_{13} = 0, g_{21} = 1, g_{22} = 1, g_{23} = 1$$

$$\begin{vmatrix} 1-z & 0 & 0 & 1 & 1 \\ 0 & 1-z & 0 & -1 & 1 \\ 0 & 0 & 1-z & 0 & 1 \\ 1 & -1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{vmatrix} = 0$$

$$6(1-z) = 0 \Rightarrow z = 1 > 0$$

$\therefore$ solution so obtained represent the relative minimum

- (1) Find the maximum of the function $f(x) = 2x_1 + x_2 + 10$ subject to $g(x) = x_1 + 2x_2^2 - 3$ using the lagrange's multiplier method.

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

FORMULATION & METHOD OF LAGRANGE'S MULTIPLIER

LECTURE NO: -23

Formulation & Method of Lagrange's Multiplier

$$L(x_1, x_2, \lambda) = f(x) + \lambda g(x)$$

$$= 2x_1 + x_2 + 10 + \lambda(x_1 + 2x_2^2 - 3)$$

Necessary condition for the extreme points are:

$$\frac{\partial L}{\partial x_1} = 2 + \lambda = 0$$

$$\frac{\partial L}{\partial x_2} = 1 + 4\lambda x_2 = 0$$

$$\frac{\partial L}{\partial \lambda} = x_1 + 2x_2^2 - 3 = 0$$

$$\Rightarrow \lambda = -2, x_2 = \frac{-1}{4(-2)} = \frac{1}{8}$$

$$\text{And } x_1 = 3 - 2x_2^2 = \frac{95}{32}$$

$$\therefore x^* = \left(\frac{95}{32}, \frac{1}{8} \right) \text{ and } \lambda = -2$$

Sufficient condition is.

$$\begin{vmatrix} L_{11} - z & L_{12} & g_{11} \\ 0 & L_{22} - z & g_{12} \\ g_{11} & g_{12} & 0 \end{vmatrix} = 0$$

$$D = \begin{vmatrix} 0 - z & 0 & 0 \\ 0 & 4\lambda - z & 4x_2 \\ 1 & 4x_2 & 6 \end{vmatrix} = 0$$

$$\text{At } x^* = \left(\frac{95}{32}, \frac{1}{8} \right) \text{ and } \lambda = -2$$

$$D = \begin{vmatrix} -z & 0 & 0 \\ 0 & -8 - z & 1/2 \\ 1 & 1/2 & 0 \end{vmatrix} = 0$$

$$\Rightarrow \frac{z}{4} + 8 + z = 0$$

$$\Rightarrow z = -6.4 < 0$$

$\therefore$ value of z is negative, x^* is a point of maxima and maximum value of f is

$$\begin{aligned} f(x^*) &= 2\left(\frac{95}{32}\right) + \frac{2}{8} + 10 \\ &= \frac{257}{16} \approx 16.0625. \end{aligned}$$

(3) Find the dimensions of a cylindrical tin (with top and bottom made u of sheet metal to maximize its volume such that the total surface area is equal to $A_0 = 24\pi$

Sol. Let x_1 and x_2 denote the radius of the base and length of the tin respectively.

The required problem is.

Maximum $f(x_1, x_2) = \pi x_1^2 x_2$

Subject to $2\pi x_1^2 + 2\pi x_1 x_2 = A_0 = 4\pi$

Define the largrange function

$$L(x, x_2, \lambda) = \pi x_1^2 x_2 + \lambda(2\pi x_1^2 + 2\pi x_1 x_2 - A_0)$$

$$\frac{\partial L}{\partial x_1} = 2\pi x_1 x_2 + 4\lambda \pi x_1 + 2\pi x_2 \lambda = 0$$

$$\frac{\partial L}{\partial x_2} = \pi x_1^2 + 2\lambda \pi x_1 = 0$$

$$\frac{\partial L}{\partial \lambda} = 2\pi x_1^2 + 2\pi x_1 x_2 - A_0 = 0$$

$$\lambda = \frac{-x_1 x_2}{2x_1 + x_2} = -\frac{1}{2} x_1 \Rightarrow x_1 = \frac{1}{2} x_2$$

$$x_1^* = \sqrt{\frac{A_0}{6\pi}}, x_2^* = \sqrt{\frac{2A_0}{3\pi}} \text{ and } \lambda^* = 1 - \sqrt{\frac{A_0}{2\pi}}$$

$$f(x_1^*, x_2^*) = \frac{A_0 \sqrt{A_0}}{\sqrt{54\pi}}$$

For $A_0 = 24\pi, x_1^* = 2, x_2^* = 4, \lambda^0 = -1$ and

$$f(x_1^*, x_2^*) = 16\pi$$

Now for the sufficient conditions.

$$L_{11} = (2\pi x_2 + 4\pi\lambda)(x^*, \lambda^*) = 4\pi$$

$$L_{12} = L_{21} = (2\pi x_1 + 2\pi\lambda)(x^*, \lambda^*) = 2\pi$$

$$L_{22} = 0$$

$$g_{11} = (4\pi x_1 + 2\pi x_2)(x^*, \lambda^*) = 16\pi$$

$$g_{12} = 2\pi x_1(x^*, \lambda^*) = 4\pi$$

$$\begin{vmatrix} 4\pi - z & 2\pi & 16\pi \\ 2\pi & 0 - z & 4\pi \\ 16\pi & 4\pi & 0 \end{vmatrix} = 0$$

$$272\pi^2 z + 192\pi^3 = 0 \Rightarrow z = \frac{-12}{17}\pi < 0$$

(4) Obtain the extreme points the function $f(x_1, x_2, x_3) = x_1 + x_2 + x_3$ subject to $x_1^2 + x_2^2 + x_3^2 = 1$ find whether the extreme points are maxima or minima.

Sol. The lagrangian function can be written as.

$$\begin{aligned} L(x, \lambda) &= f(x) + \lambda g(x) \\ &= x_1 + x_2 + x_3 + \lambda(x_1^2 + x_2^2 + x_3^2 - 1) \end{aligned}$$

Necessary conditions for the points of maxima of minima are.

$$\frac{\partial L}{\partial x_1} = 0 \Rightarrow 1 = 2x_1\lambda = 0 \dots\dots\dots(1)$$

$$\frac{\partial L}{\partial x_2} = 0 \Rightarrow 1 = 2x_2\lambda = 0 \dots\dots\dots(2)$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow 1 = 2x_3\lambda = 0 \dots\dots\dots(3)$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow x_1^2 + x_2^2 + x_3^2 - 1 = 0 \dots\dots\dots(4)$$

Solving (1),(ii),(iii) & (iv), we obtain.

$$x_1 = x_2 = x_3 = -1/2\lambda$$

$$\Rightarrow \frac{3}{4\lambda^2} - 1 = 0 \Rightarrow \lambda = \pm \frac{\sqrt{3}}{2}$$

$$\Rightarrow x_1^* \left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{\sqrt{1}}{\sqrt{3}} \right) \text{ and } x_2^* \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

are the extreme points for sufficient condition, solving the determinant

$$\begin{vmatrix} L_{11} - z & L_{12} & L_{13} & g_{11} \\ L_{21} & L_{22} - z & L_{23} & g_{12} \\ L_{31} & L_{32} & L_{33} - z & g_{13} \\ g_{11} & g_{12} & g_{13} & 0 \end{vmatrix} = 0$$

Where $L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j}$ and $g_{ij} = \frac{\partial g_i}{\partial x_j}$

$$\begin{vmatrix} 2\pi - z & 0 & 0 & 2x_1 \\ 0 & 2\lambda - z & 0 & 2x_2 \\ 0 & 0 & 2\lambda - z & 2x_3 \\ 2x_1 & 2x_2 & 2x_3 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (2\lambda - z) \begin{vmatrix} 2\lambda - z & 0 & 2x_2 \\ 0 & 2\lambda - z & 2x_3 \\ 2x_2 & 2x_3 & 0 \end{vmatrix} - x_1 \begin{vmatrix} 0 & 2\lambda - z & 0 \\ 0 & 0 & 2\lambda - z \\ 2x_2 & 2x_2 & 2x_3 \end{vmatrix} = 0$$

$$\Rightarrow (2\lambda - z)^2 (-4x_3^2 - 4x_2^2) - (2\lambda - z)^2 (4x_1^2) = 0$$

$$\Rightarrow (2\lambda - z)(x_1^2 + x_2^2 + x_3^2) = 0$$

$$\Rightarrow z = 2\lambda, 2\lambda$$

Therefore.

(i) If $\lambda = \frac{\sqrt{3}}{2}, z > 0 \Rightarrow x_1^* \left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right)$ is point of minima.

(ii)

(iii) If $\lambda = \frac{-\sqrt{3}}{2}, z > 0 \Rightarrow x_2^* \left(\frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right)$ is a point of maxima.

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

KUHN TUCKER CONDITIONS(MULTIVARIABLE OPTIMIZATION WITH INEQUALITY CONSTRAINTS)

LECTURE NO: - 24

Kuhn Tucker conditions(Multivariable optimization with inequality constraints).

If the problem is.

Minimize $f(x)$

Subject to

$$\left. \begin{array}{l} g_j(x) \leq 0 \quad j = 1, 2, \dots, m \\ x(x_1, x_2, \dots, x_n)^T \quad m < n \end{array} \right\} \dots \dots \dots (2)$$

Then at a constrained minimum point x^* the following conditions must be satisfied.

$$\left. \begin{array}{l} \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad i = 1, 2, \dots, n \\ \lambda_j g_j = 0 \quad j = 1, 2, \dots, m \\ g_j \leq 0 \quad j = 1, 2, \dots, m \\ \lambda_j \geq 0 \quad j = 1, 2, \dots, m \end{array} \right\} \dots \dots \dots (2)$$

These are called the Kuhn-Tucker condition after the mathematician who derived them.

- (1) Kuhn-Tucker conditions are only necessary conditions to be satisfied at a relative minimum of $f(x)$
- (2) If the problem is a convex programming problem then these conditions are sufficient also.
- (3) If the problem is of maximization i.e.

Maximize $f(x)$

Subject to $g_j(x) \leq 0 \quad j = 1, 2, \dots, m$

Or If the constraints are $g_j(x) \geq 0 \quad j = 1, 2, \dots, m$

Minimize $f(x)$

Subject to $g_j(x) \geq 0 \quad j = 1, 2, \dots, m$

Then in Kuhn-Tucker conditions.

$$\lambda_j \leq 0 \quad j = 1, 2, \dots, m$$

On the other hand if the problem is of maximization and $g_j(x) \geq 0, j = 1, 2, \dots, m$ the conditions remain the same.

Now we state the problem with both equality and inequality constraints.

$$\left. \begin{array}{l} \text{minimize } f(x) \\ \text{subject to } g_j(x) \leq 0, \quad j = 1, 2, \dots, m \\ \quad \quad \quad h_k(x) = 0 \quad k = 1, 2, \dots, p \end{array} \right\} \dots \dots (3)$$

The lagrange function is.

$$L(x, \lambda, \mu) = f(x) + \sum_{j=1}^m \lambda_j g_j(x) + \sum_{k=1}^p \mu_k h_k(x)$$

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)^T, \mu = (\mu_1, \mu_2, \dots, \mu_p)^T$$

Are lagrange multipliers.

The Kuhn-Tucker conditions become.

$$\left. \begin{array}{l} \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} + \sum_{k=1}^p \mu_k \frac{\partial h_k}{\partial x_i} = 0 \quad i = 1, 2, \dots, n \\ \lambda_j g_j = 0 \quad j = 1, 2, \dots, m \\ g_j \leq 0 \quad j = 1, 2, \dots, m \\ h_k = 0 \quad k = 1, 2, \dots, p \\ \lambda_j \geq 0 \quad j = 1, 2, \dots, m \end{array} \right\} \dots \dots (4)$$

Constraint Qualification:-

Let x^* be a feasible solution to the problem (3) of if $\nabla g_j(x^*), j \in J_1$ (Set of active constraints) and $\nabla h_k(x^*), k = 1, 2, \dots, p$ are linearly independent, there exist λ^* and μ^* such that (x^*, λ^*, μ^*) satisfy equation (4)"

The requirement $\nabla g_j(x^*), j \in J_1$ and $\nabla h_k(x^*), k = 1, 2, \dots, p$ be linearly independent is called the constraint qualification.

Loosly put the constraint qualification states that there is a feasible solution that lies strictly in the interior of the region defined by $g_j(x) \leq 0, j = 1, 2, \dots, m$

If the constraint qualifications are not satisfied at (x^*, λ^*, μ^*) , (4) may or may not have solution it is difficult to verify the constraint qualification unless we know. x^* For a problem having any of the following properties, constraint qualifications are always satisfied.

- 1- All constraints (inequality and equality) are linear
- 2- All equalities are linear and inequality constraint functions are convex and at least one feasible vector lies strictly inside the feasible region so that.

$$gj(\hat{x}) < 0, j = 1, 2, \dots, m; hk(\hat{x}) = 0, k = 1, 2, \dots, p$$

($\hat{x}$ is the feasible vector)

(1) Consider the following optimization problem.

$$\text{Maximize } f = -x_1 - x_2$$

$$\text{Subject to } x_1^2 + x_2 \geq 3x_2$$

$$x_1 + x_2^4 \leq 30$$

(a) Find whether the vector $x^* = (1, 1)^T$ satisfies the Kuhn-Tucker conditions for a constrained optimum

(b) What are the value of the lagrange multiplies at the given vector

Sol. The given problem can be written as.

$$\text{Maximize } f = -x_1 - x_2$$

Subject to

$$x_1^2 + x_2 - 2 \geq 0$$

$$x_1 + 3x_2 - 4 \geq 0$$

$$-x_1 - x_2^4 + 30 \geq 0$$

Define the lagrange function

$$L(x, \lambda) = -x_1 - x_2 + \lambda_1(x_1^2 + x_2 - 2) + \lambda_2(x_1 + 3x_2 - 4) + \lambda_3(-x_1 - x_2^4 + 30)$$

$$= f(x_1, x_2) + \lambda_1 g_1(x_1, x_2) + \lambda_2 g_2(x_1, x_2) + \lambda_3 g_3(x_1, x_2)$$

Kum-Tacker conditions are :

$$\frac{\partial L}{\partial x_1} = -1 + 2\lambda_1 x_1 = \lambda_2 - \lambda_3 = 0$$

$$\frac{\partial L}{\partial x_2} = -1 + \lambda_1 + 3\lambda_2 - 4\lambda_3 x_2^3 = 0$$

$$\lambda_j g_j = 0 \quad j = 1, 2, 3$$

$$gj(x_1, x_2) \geq 0 \quad j = 1, 2, 3$$

$$\lambda_j \geq 0 \quad j = 1, 2, 3$$

For $x^* = (1, 1)^T$

$$g_1(x^*) = 1 + 1 - 2 = 0$$

$$g_2(x^*) = 1 + 3 - 4 = 0$$

$$g_3(x^*) = -11 - +30 - 28 \Rightarrow \lambda_3 = 0$$

First two K.T. conditions gives (at x^*)

$$\begin{cases} -1 + 2\lambda_1 + \lambda_2 = 0 \\ -1 + \lambda_1 + 3\lambda_2 = 0 \end{cases} \Rightarrow \lambda_1^* = \frac{2}{5}, \lambda_2^* = \frac{1}{5}$$

Gives $\lambda_1^*, \lambda_2^* > 0, \lambda_3^* = 0$

(a) Hence vector $x^* = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ satisfies the K.T. conditions

$$(b) \lambda_1^* = \frac{2}{5}, \lambda_2^* = \frac{1}{5}, \lambda_3^* = 0$$

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors.

UNIT-3 (ADVANCE ENGINEERING MATHEMATICS)

PROBLEMS ON KUHN TUCKER CONDITIONS

LECTURE NO: - 25

Problems on Kuhn Tucker conditions

If the problem is.

Minimize $f(x)$

Subject to

$$\left. \begin{aligned} g_j(x) &\leq 0 \quad j = 1, 2, \dots, m \\ x(x_1, x_2, \dots, x_n)^T \quad m < n \end{aligned} \right\} \dots\dots\dots (2)$$

Then at a constrained minimum point x^* the following conditions must be satisfied.

$$\left. \begin{aligned} \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} &= 0, \quad i = 1, 2, \dots, n \\ \lambda_j g_j &= 0 \quad j = 1, 2, \dots, m \\ g_j &\leq 0 \quad j = 1, 2, \dots, m \\ \lambda_j &\geq 0 \quad j = 1, 2, \dots, m \end{aligned} \right\} \dots\dots\dots (2)$$

These are called the Kuhn-Tucker condition after the mathematician who derived them.

- (1) Kuhn-Tucker conditions are only necessary conditions to be satisfied at a relative minimum of $f(x)$
- (2) If the problem is a convex programming problem then these conditions are sufficient also.
- (3) If the problem is of maximization i.e.

Maximize $f(x)$

Subject to $g_j(x) \leq 0 \quad j = 1, 2, \dots, m$

Or If the constraints are $g_j(x) \geq 0 \quad j = 1, 2, \dots, m$

Minimize $f(x)$

Subject to $g_j(x) \geq 0 \quad j = 1, 2, \dots, m$

Then in Kuhn-Tucker conditions.

$$\lambda_j \leq 0 \quad j = 1, 2, \dots, m$$

On the other hand if the problem is of maximization and $g_j(x) \geq 0, j = 1, 2, \dots, m$ the conditions remain the same.

Now we state the problem with both equality and inequality constraints.

$$\left. \begin{array}{l} \text{minimize } f(x) \\ \text{subject to } g_j(x) \leq 0, \quad j = 1, 2, \dots, m \\ \quad \quad \quad h_k(x) = 0 \quad k = 1, 2, \dots, p \end{array} \right\} \dots \dots (3)$$

The lagrange function is.

$$L(x, \lambda, \mu) = f(x) + \sum_{j=1}^m \lambda_j g_j(x) + \sum_{k=1}^p \mu_k h_k(x)$$

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)^T, \mu = (\mu_1, \mu_2, \dots, \mu_p)^T$$

Are lagrange multipliers.

The Kuhn-Tucker conditions become.

$$\left. \begin{array}{l} \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} + \sum_{k=1}^p \mu_k \frac{\partial h_k}{\partial x_i} = 0 \quad i = 1, 2, \dots, n \\ \lambda_j g_j = 0 \quad j = 1, 2, \dots, m \\ g_j \leq 0 \quad j = 1, 2, \dots, m \\ h_k = 0 \quad k = 1, 2, \dots, p \\ \lambda_j \geq 0 \quad j = 1, 2, \dots, m \end{array} \right\} \dots \dots (4)$$

Constraint Qualification:-

Let x^* be a feasible solution to the problem (3) of if $\nabla g_j(x^*), j \in J_1$ (Set of active constraints) and $\nabla h_k(x^*), k = 1, 2, \dots, p$ are linearly independent, there exist λ^* and μ^* such that (x^*, λ^*, μ^*) satisfy equation (4)"

The requirement $\nabla g_j(x^*), j \in J_1$ and $\nabla h_k(x^*), k = 1, 2, \dots, p$ be linearly independent is called the constraint qualification.

Loosly put the constraint qualification states that there is a feasible solution that lies strictly in the interior of the region defined by $g_j(x) \leq 0, j = 1, 2, \dots, m$

If the constraint qualifications are not satisfied at (x^*, λ^*, μ^*) , (4) may or may not have solution it is difficult to verify the constraint qualification unless we know. x^* For a problem having any of the following properties, constraint qualifications are always satisfied.

- 1- All constraints (inequality and equality) are linear
- 2- All equalities are linear and inequality constraint functions are convex and at least one feasible vector lies strictly inside the feasible region so that.
 $g_j(\hat{x}) < 0, j = 1, 2, \dots, m; h_k(\hat{x}) = 0, k = 1, 2, \dots, p$

($\hat{x}$ is the feasible vector)

(1) Consider the following optimization problem.

$$\text{Maximize } f = -x_1 - x_2$$

$$\text{Subject to } x_1^2 + x_2 \geq 3x_2$$

$$x_1 + x_2^4 \leq 30$$

(a) Find whether the vector $x^* = (1, 1)^T$ satisfies the Kuhn-Tucker conditions for a constrained optimum

(b) What are the value of the lagrange multiplies at the given vector

Sol. The given problem can be written as.

$$\text{Maximize } f = -x_1 - x_2$$

Subject to

$$x_1^2 + x_2 - 2 \geq 0$$

$$x_1 + 3x_2 - 4 \geq 0$$

$$-x_1 - x_2^4 + 30 \geq 0$$

Define the lagrange function

$$L(x, \lambda) = -x_1 - x_2 + \lambda_1(x_1^2 + x_2 - 2) + \lambda_2(x_1 + 3x_2 - 4) + \lambda_3(-x_1 - x_2^4 + 30)$$

$$= f(x_1, x_2) + \lambda_1 g_1(x_1, x_2) + \lambda_2 g_2(x_1, x_2) + \lambda_3 g_3(x_1, x_2)$$

Kum-Tacker conditions are :

$$\frac{\partial L}{\partial x_1} = -1 + 2\lambda_1 x_1 = \lambda_2 - \lambda_3 = 0$$

$$\frac{\partial L}{\partial x_2} = -1 + \lambda_1 + 3\lambda_2 - 4\lambda_3 x_2^3 = 0$$

$$\lambda_j g_j = 0 \quad j = 1, 2, 3$$

$$g_j(x_1, x_2) \geq 0 \quad j = 1, 2, 3$$

$$\lambda_j \geq 0 \quad j = 1, 2, 3$$

For $x^* = (1, 1)^T$

$$g_1(x^*) = 1 + 1 - 2 = 0$$

$$g_2(x^*) = 1 + 3 - 4 = 0$$

$$g_3(x^*) = -11 - +30 - 28 \Rightarrow \lambda_3 = 0$$

First two K.T. conditions gives (at x^*)

$$\begin{cases} -1 + 2\lambda_1 + \lambda_2 = 0 \\ -1 + \lambda_1 + 3\lambda_2 = 0 \end{cases} \Rightarrow \lambda_1^* = \frac{2}{5}, \lambda_2^* = \frac{1}{5}$$

Gives $\lambda_1^*, \lambda_2^* > 0, \lambda_3^* = 0$

(a) Hence vector $x^* = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ satisfies the K.T. conditions

$$(b) \lambda_1^* = \frac{2}{5}, \lambda_2^* = \frac{1}{5}, \lambda_3^* = 0$$

References:

Chandrika Prasad, Advance Engineering Mathematics, Standard Publishers and Distributors.